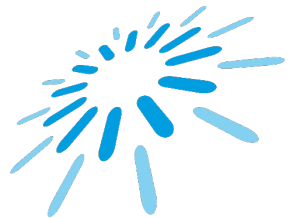




Collaboration, Quality, Test

Prof. Dr. Jóakim von Kistowski



TH Aschaffenburg
university of applied sciences

Check-In, Part I

Your key learnings this semester (so far)?

Organization

Prof. Dr. Jóakim von Kistowski



Computer Science

Karlsruhe Institute of Technology

PhD Computer Science

University of Würzburg

Research Group Leader

University of Würzburg

Senior Software Architect / Team Lead

DATEV eG

Member, Chair, Contributor

Standard Performance Evaluation Corporation (SPEC)



Research Topics

Looking for Bachelor/Master Thesis Topics?

GreenIT and Energy Efficiency

- Energy Efficiency of Servers
- Energy Efficiency of Software
- Green Software Operations
- Efficient Management of Server Fleets

Software Architecture and Quality

- Microservices Architectures
- Cloud-native Computing
- Software Resilience and Chaos Testing
- Software Performance Benchmarking and Analysis

Creating an interactive Rally Tour Guide for a handheld gaming device

Context:

- We want to hand out interactive tour guides on events such as open campuses
- These devices should be cheap
- We are targeting devices with aarch64 CPUs with linux operation systems
- Challenge: No GPS on the devices

Goal:

- Solve the technical challenges of getting an interactive tour guide incl. location tracking working on an aarch64 linux device without any GPS



	Monday	Tuesday	Wednesday	Thursday	Friday
08:00 - 09:30					
09:45 - 11:15				Vorlesung Software Design	
11:30 - 13:00			Lecture Software Design International	Übung Software Design	
	Lunch Break				
14:00 - 15:30			Exercises (Group 2) Software Design International	Exercises (Group 1) Software Design International	
15:45 - 17:15					
17:30 - 19:00					

Lectures and Practical Exercises

Lectures

- Room:
SD: C2-225
SDI: C2-302
- Totally interactive!

Practical Exercises

- Room:
SD: C2-127
SDI Group 1: C2-226
SDI Group 2: C2-225
- In small groups
- Bring your PC
- Pen and some paper helps
- Programming exercises and worksheets

Moodle Course

German:

- WiSe2526_KQT

International:

- WiSe2526_CQT_SDI

- Enrollment Key: **NegativeExponentialDistribution_2019**

Check-In, Part II

What do you expect from this lecture?

Agenda



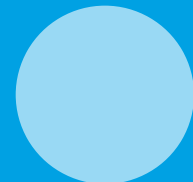
Organization

Recap

Certified Tester Foundation Level

Software Errors

Testing Things



Why do we have to deal with software quality?

Question

What are the challenges of software quality assurance?

How do we ensure software quality?

How do we ensure **Quality**?

- **Quality management:** established process for dealing with quality
 - How can quality be described? (→ Requirements Engineering)
 - How can the quality of software (SW) be assessed? (→ SW testing)
 - How can quality be achieved? (→ entire lecture SWE, professional software engineering)
 - How to find errors/defects? (→ SW testing)
 - How to avoid errors/defects? (→ entire lecture SWE, professional software engineering)
- **Quality assurance (QA):** concrete procedure for ensuring quality
 - Constructive QA
 - Analytical QA
 - Organizational QA

Organizational

Scan Health
Insurance Card

Go to your
medical check-up

Open a
practice

Analytic

Place info
brochures in the
waiting room

Make an
appointment

Constructive

Create an
advanced training
program for
doctors

Examine
the patient

Draw blood

Organizational

Scan Health
Insurance Card

Make an
appointment

Open a
practice

Analytic

Draw blood

Examine
the patient

Constructive

Create an
advanced training
program for
doctors

Place info
brochures in the
waiting room

Go to your
medical check-up

Organizational

Scan Health Insurance Card

Organization

Make an appointment

Open a practice

Other names for the categories?

Analytic

Diagnosis

Draw blood

Examine the patient

Constructive

Prevention

Create an advanced training program for doctors

Go to your medical check-up

Place info brochures in the waiting room

Organizational

Organization

Analytic

Diagnosis

Constructive

Prevention

What methods do you
know from SWE?

Organizational

Organization

Roles, Milestones

Project
management

Analytic

Diagnosis

Test

Reviews

Constructive

Prevention

Architecture
training

Requirements
engineering

What methods do you
know from SWE?

Software Quality Assurance

Organizational

Organization

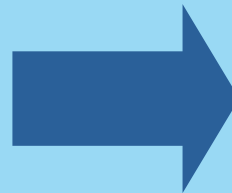
Analytic

Diagnosis

Constructive

Prevention

Quality cannot
be tested
„into a product“!



Organizational and constructive measures are significantly more important, they are supplemented by analytical measures.

Software Quality Assurance

Organizational

Organization

Analytic

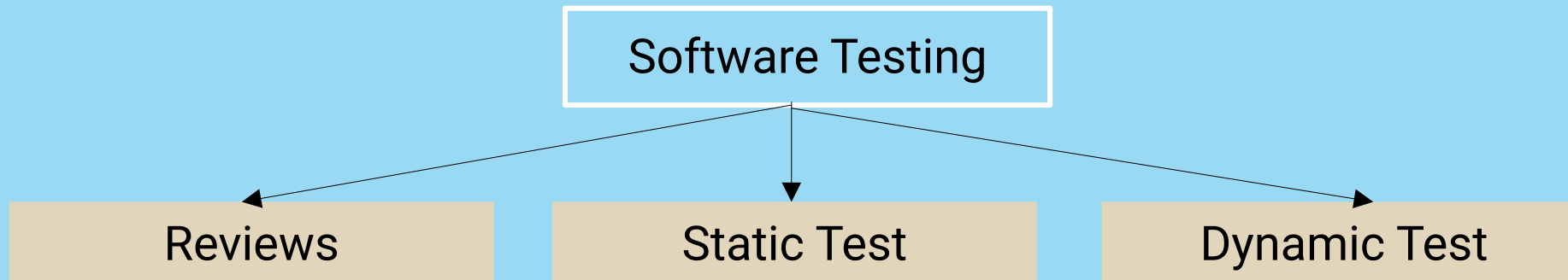
Diagnosis

Constructive

Prevention

Software project management

constructive
Software Engineering



Software Quality Assurance

Organizational

Organization

Analytic

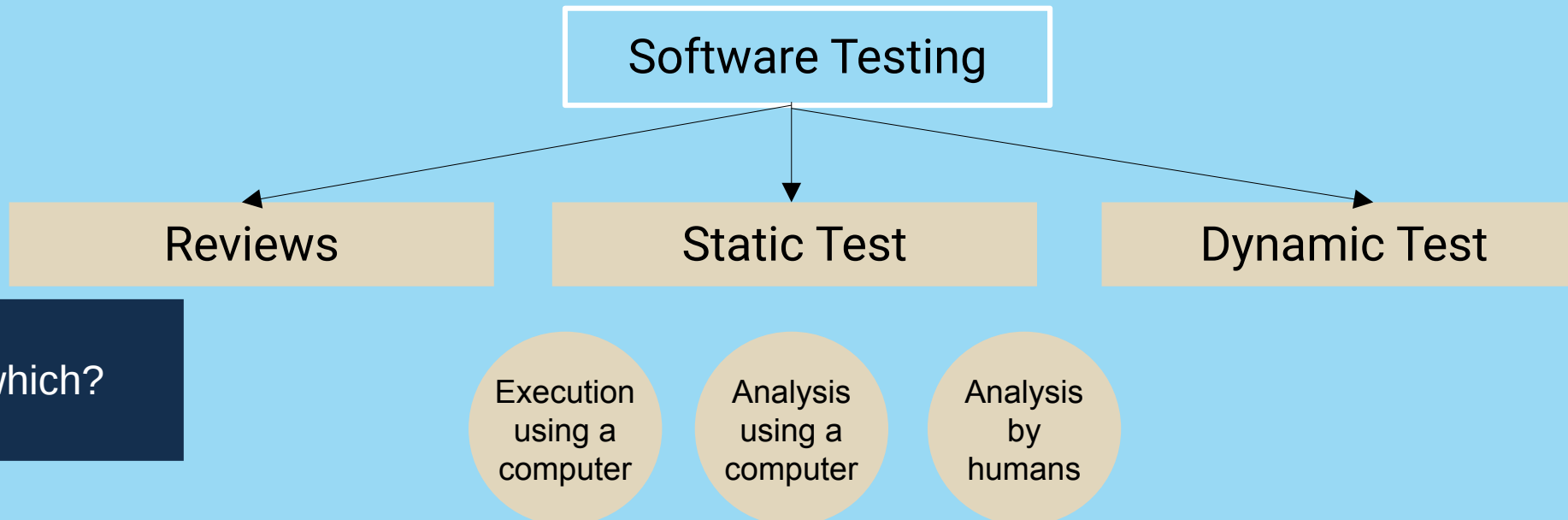
Diagnosis

Constructive

Prevention

Software project management

constructive
Software Engineering



Which is which?

Software Quality Assurance

Organizational

Organization

Analytic

Diagnosis

Constructive

Prevention

Software project management

constructive
Software Engineering

Software Testing

Reviews

Static Test

Dynamic Test

Which is which?

Analysis
by
humans

Analysis
using a
computer

Execution
using a
computer

Agenda

Organization

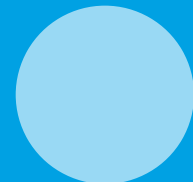
Recap



Certified Tester Foundation Level

Software Errors

Testing Things



Software Quality Assurance

Organizational

Organization

Analytic

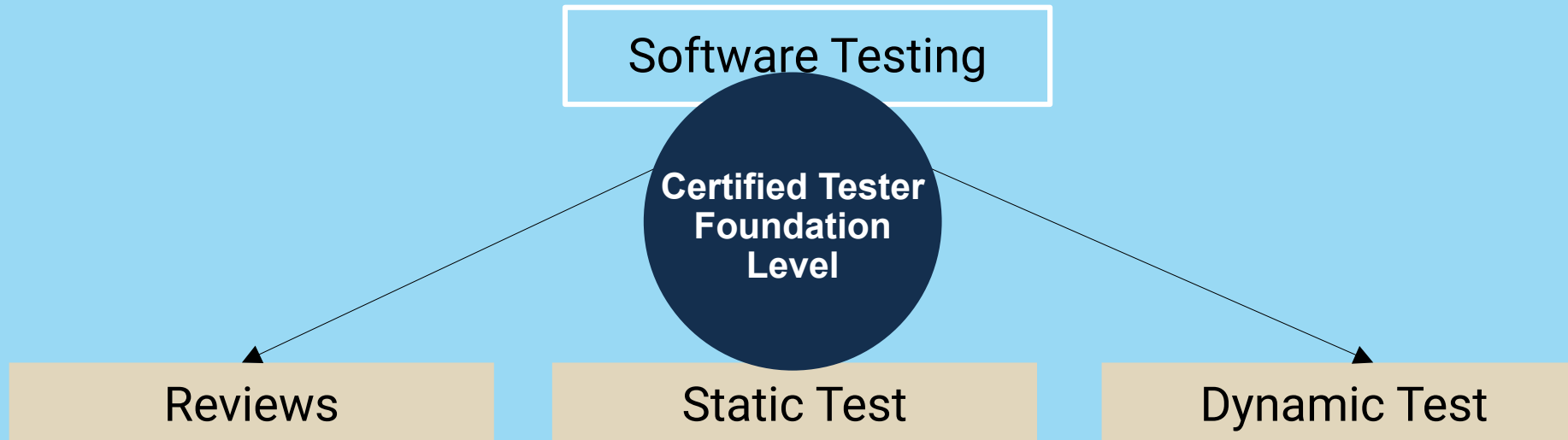
Diagnosis

Constructive

Prevention

Software project management

constructive
Software Engineering



About this Lecture



Loosely based on Certified Tester Course

- Class is based on a course designed by the German Testing Board (GTB)
- May be used as a basis for a later participation in the certification program
- **Warning:** This class deviates from the GTB Syllabus in some aspects!

Thanks and Acknowledgments

A large part of the slides were taken from a joint **cross-university course** and updated by the **GTB** in accordance with the current updated according to the current German-language curriculum V4.0.



- The initial version for TH Aschaffenburg was created by
 - **Prof. Dr. Timea Illes-Seiffert**
- The original GTB version was created by
 - Dr. Falk Fraikin, TU Darmstadt
 - Dr. Eike Hagen Riedemann, ehemals TU Dortmund
 - Prof. Dr. Andreas Spillner, Hochschule Bremen
 - Prof. Dr. Mario Winter, FH Köln
- **Warning:** This class deviates from the GTB Syllabus in some aspects!

Important!

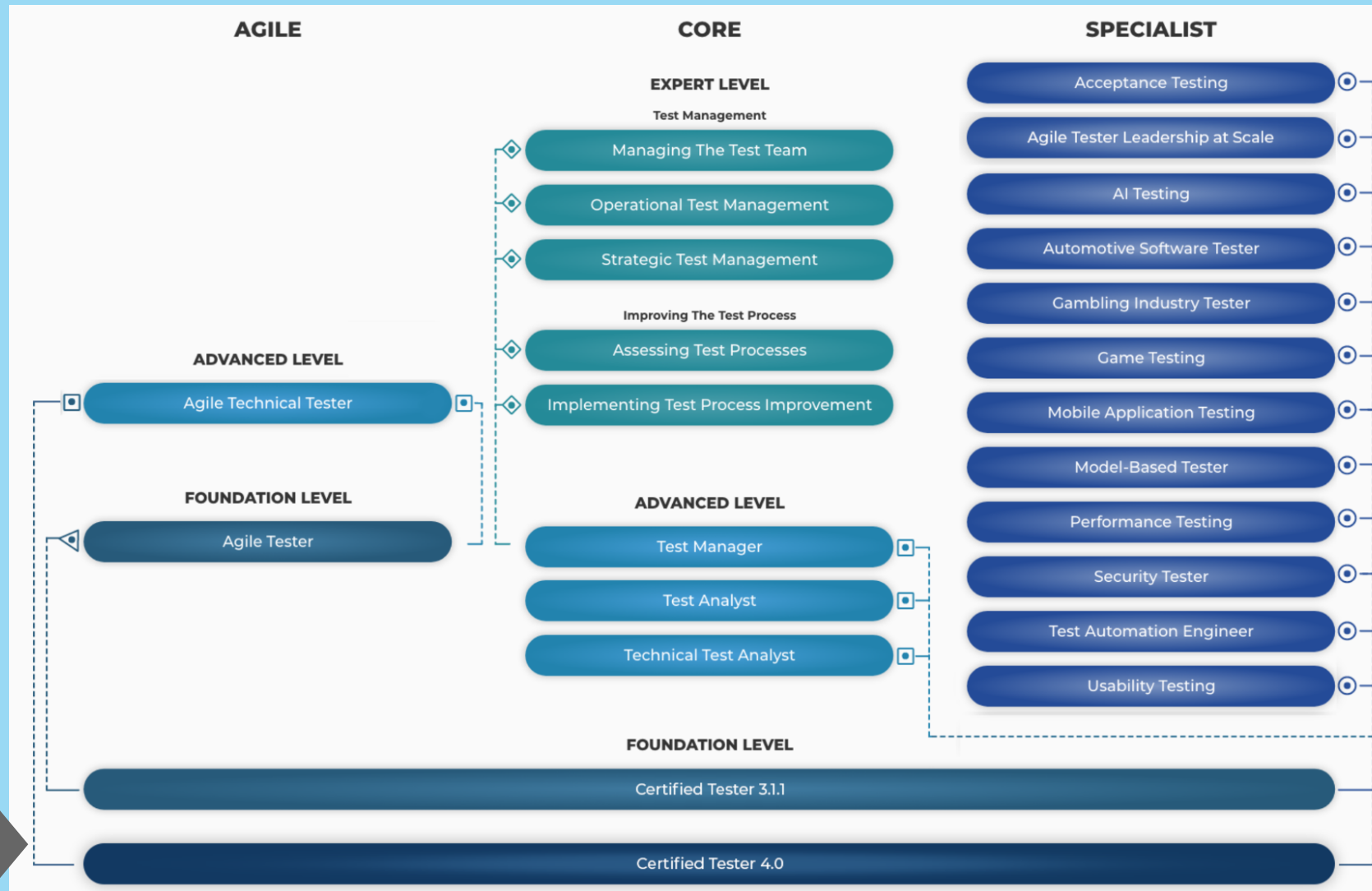
Free distribution of these slides
on the Internet or commercial use is
not permitted!

Research the Certified Tester Foundation Level (CTFL).

What did you find?

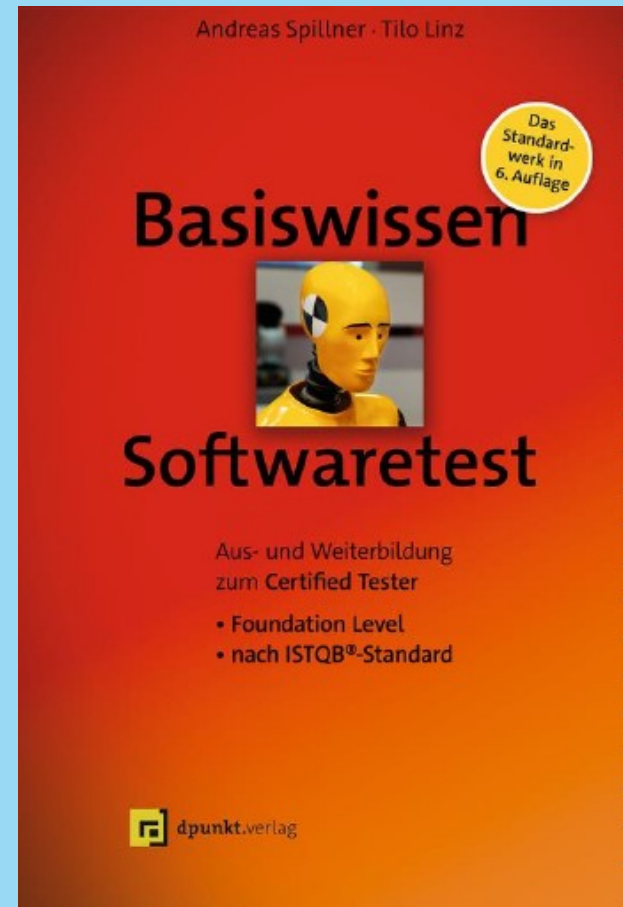
Goals for this Class

- Learning the fundamentals of quality assurance, examining and testing software
- Explanation of terms, tasks and activities, methods and test design procedures
- The content of the course is based on an internationally defined teaching material for higher education institutions

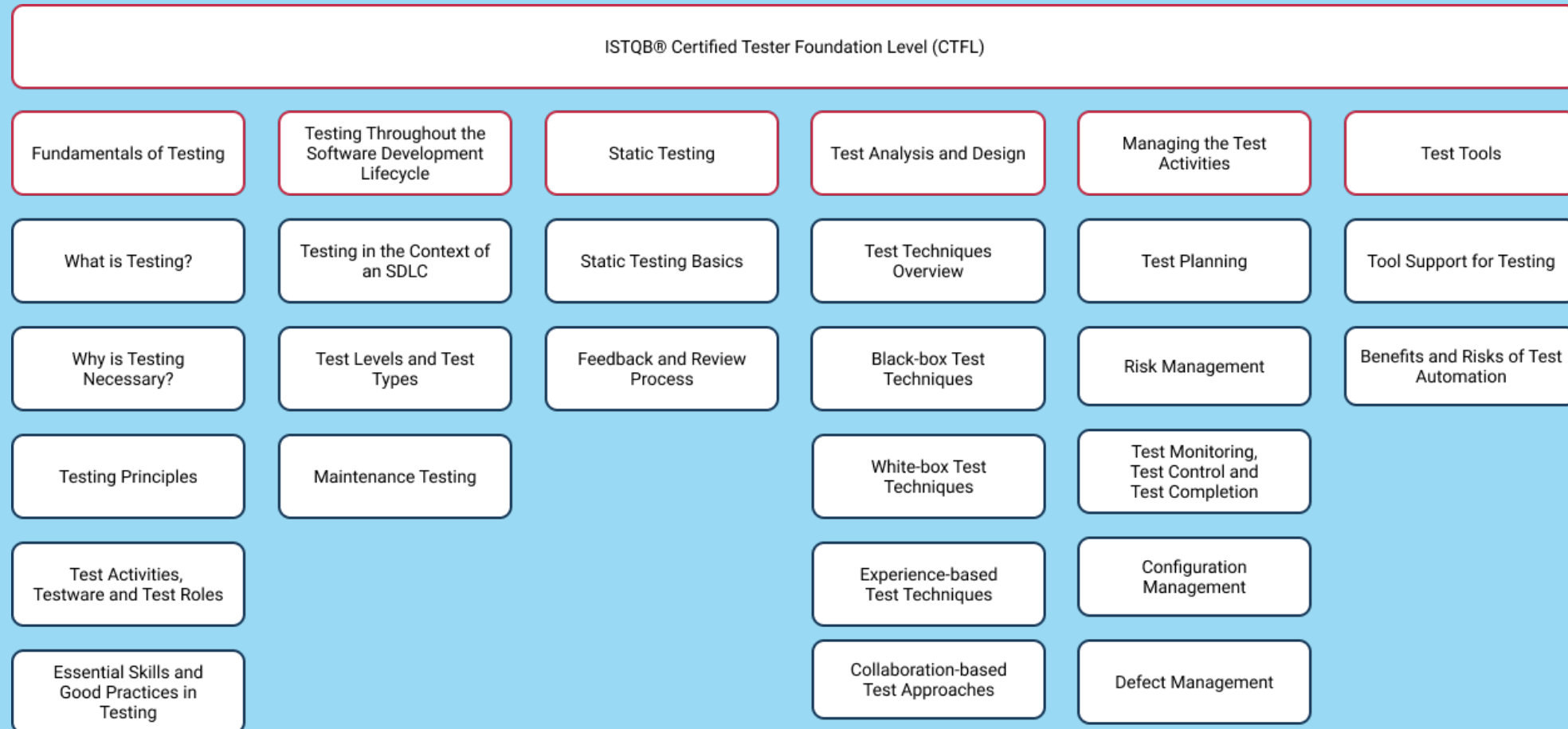


Book for Certified Tester – Foundation Level (German)

- Andreas Spillner, Tilo Linz:
Basiswissen Softwaretest
Aus- und Weiterbildung zum Certified Tester
Foundation Level nach ISTQB-Standard
 - 6., überarbeitete Auflage
 - Ausgabe 12. Juni 2019
 - Preis: 39,90 EUR
 - 378 Seiten
 - ISBN Print: 978-3-86490-583-4
 - ISBN PDF: 978-3-96088-501-6
 - ISBN ePub: 978-3-96088-502-3
 - ISBN Mobi: 978-3-96088-503-0
- -dpunkt.verlag: <http://www.dpunkt.de/>



**EBook available
in the Library**

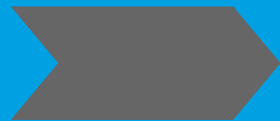


Agenda

Organization

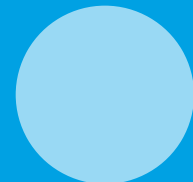
Recap

Certified Tester Foundation Level



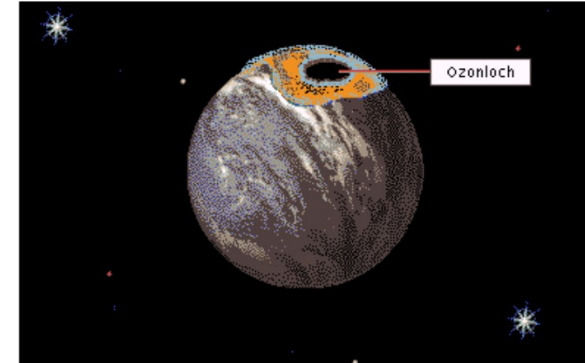
Software Errors

Testing Things

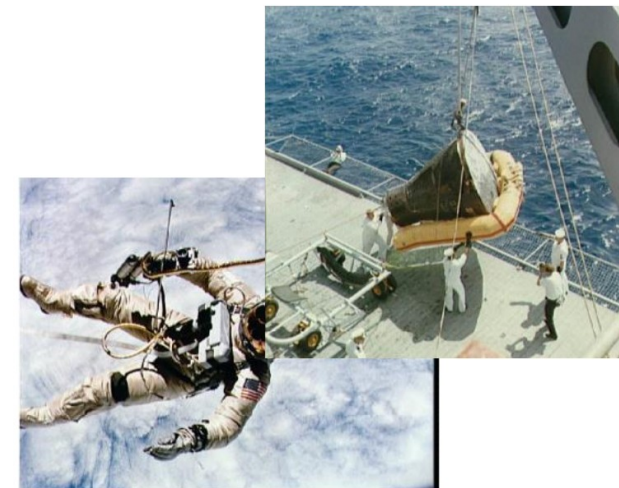


Effects of Software Errors

- **NASA - Earth observation satellites 1979-1985**
 - Ozone hole not detected for 7 years (!)
 - Cause: software error
- **ESA, Kourou, French Guyana, June 4, 1996**
 - Self-destruction of Ariane 5 during maiden flight 39 seconds after launch
 - Cause: Software error - attitude control software from Ariane 4 reused for launch trajectory of Ariane 5 without testing, resulting in **conversion error**
- **Manned NASA space capsule Gemini V**
 - Missed its landing site by approx. 160 kilometers
 - Cause: software error - rotation of the earth around the sun not taken into account!



Ariane 5 Flight 501



Task

Discuss in small groups!

Write down the **Causes** and **Effects** (damages) for the given examples.

Scan the QR Codes



T-Mobile Germany 2009

- April 21, 2009: Software error paralyzes T-Mobile network
 - Cell phones of millions of T-Mobile customers unreachable as of 4 p.m.
 - Automated announcement:

“This line is temporarily unavailable for technical reasons. Please call back later.”
 - 7 p.m.: Technicians restart the system
 - 10 p.m.: Network works again across the board
 - Cause: Software error in the Home Location Register (HLR). This is where telephone numbers are assigned to SIM cards.



Car Troubles

Alles über Audi A4

Übersicht Tests News Erbkönige Gebrauchtwagen Usermeinungen Videos Neuwagen



26 Bilder

Audi ruft 850.000 Autos zurück

Gefährlicher Airbag-Fehler im Audi A4

In diesem Artikel

1. Artikel

2. Bildergalerie

Die Steuergeräte der Frontairbags von Audis Mittelklassemodell A4 sind falsch programmiert. Mögliche Folge: Unfälle ohne Airbag-Auslösung. Die betroffenen A4 ab dem Facelift 2012 ruft Audi in die Werkstätten.

Alles über Opel Insignia

Übersicht Tests News Erbkönige Gebrauchtwagen Usermeinungen Videos Neuwagen



37 Bilder

Opel Insignia Rückruf

61.000 Fahrzeuge mit Softwarefehler

In diesem Artikel

1. Artikel

2. Bildergalerie

Opel beordert in Deutschland 61.000 Opel Insignia der Baujahre 2009 bis 2012 in die Werkstätten. Ein Softwarefehler kann zu einem Leistungsabfall oder sogar Motorschaden führen.

Further Software Errors

Fehler in der Software

Zehntausenden Studenten droht Bafög-Verspätung

Seit dem ersten August gibt es höhere Bafög-Sätze und Freibeträge. Doch eine gängige Bearbeitungs-Software kennt diese Neuerungen nicht. Deshalb könnten viele Studenten zum Semesterstart erst einmal ganz ohne Geld dastehen.

24.08.2016

f Teilen

Twittern

Teilen

E-mailen



Softwarefehler: Deutsche-Bank-Kunden kommen nur eingeschränkt an Geld

Softwarefehler

UPDATE 03.06.2016 18:20

Deutsche-Bank-Kunden kommen nur eingeschränkt an Geld

Nachdem bei der Deutschen Dank online fehlerhafte Doppelbuchungen aufgetreten sind, können Kunden vielfach keine Terminals nutzen. VON SARAH KRAMER



<https://www.tagesspiegel.de/wirtschaft/deutsche-bank-kunden-kommen-nur-eingeschraenkt-an-geld-6881890.html> am 10.02.2020

Scan the QR Codes



T-Mobile outage

Cause: Backend software error causing network congestion

Effect: Nationwide outage (USA)



Crowdstrike error

Cause: Faulty kernel module

Effect: Millions of Windows systems unbootable



Verizon billing error

Cause: Programming error

Effect: Billing plans are inadvertently charged



GM brake warning

Cause: Unspecified software error

Effect: Brake fluid warning light fails to turn on

Task

As a small group:

Find three additional software errors in Media, write down your **source**, the **Causes** and **Effects** (damages)

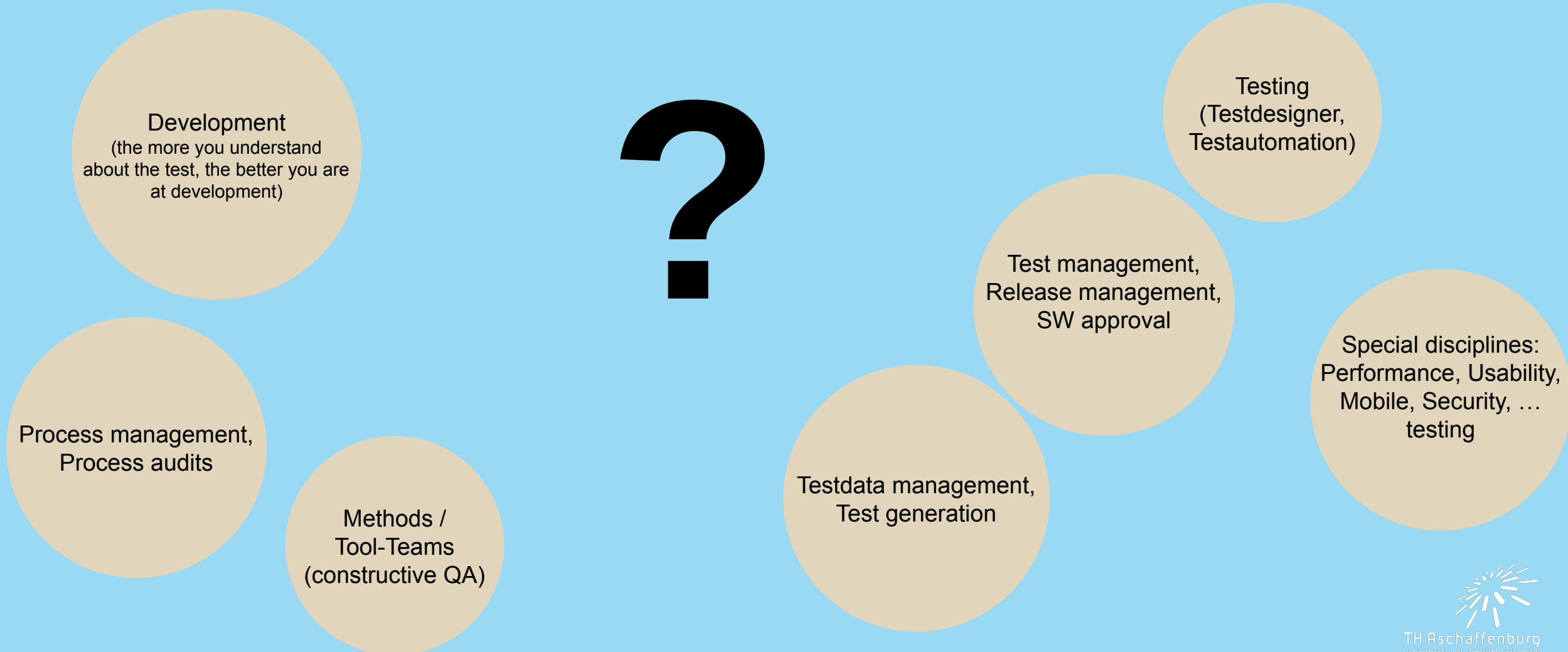
Question

What does this mean for you?

What do you do in the area of quality management and testing in industry?



What do you do in the area of quality management and testing in industry?

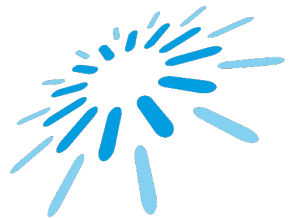


- Your next steps:**
- 1. Download Syllabus ISTQB CTFL**
 - 2. German Speakers: Get the Basiswissen Softwaretest eBook**



Until next time!

Collaboration, Quality, Test
Prof. Dr. Jóakim von Kistowski



TH Aschaffenburg
university of applied sciences